



Deep Learning attempts to mimic the activity of the neuron layers in the neocortex.

The software utilised in deep learning, in the actual true sense of the word, ‘learns’ to identify patterns in digital representations, be it audio-visual or textual, sound and images. However, success hasn’t always been achieved in these trials. Today we

have a completely different tale to tell since enhanced mathematical formulae and infinitely more powerful computers allow computer engineers and scientists to create software that can accurately model way more artificial neuron layers than ever possible before.

Why ‘deep’ learning

The origin of the name ‘deep learning’, which is a noted sub-set of machine learning has always been pondered upon and debated. To understand the reason behind the nomenclature, you will need to delve deeper into neural networks. Chapter O3 will explain the concept comprehensively, however, this is a basic explanation to warrant the name ‘deep learning’. Neural networks are basically mathematical algorithms or codes that are loosely based on the neurons in the human brain.

Let’s look at a single neuron in a neural network as a language translator in an embassy. She takes in data or ‘words’ spoken by a political leader via an input, in this case, her ears, transforms the data to a more compliant format, in this case, the language of the translator, and throws it out via an output, here it could be in spoken words format or a written document. To understand a neural network, you will simply need to multiply this one neuron or ‘translator’ into multiple and all of them are arranged in different layers.

Basically, in ‘deep learning’ hundreds of hidden layers of Artificial Neural Networks are present. The multiple concealed layers of neurons are sandwiched between the input and output layers of the software. Each concealed layer, ultimately, feeds data to each other, which allows the kind of digital and data representations and predictions that are not possible by ‘Shallow machine learning’. This is why the term ‘deep’ is used in ‘deep’ learning and ‘deep’ neural networks. It essentially refers to a large number